What Can Machine Learning Predict in NASDAQ Markets?

Price, Direction and Volatility Forecasting Across Models and Timeframes

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AI, Machine Learning and Big Data for Finance and Management · Waseda University
Final report · November 2026 · E-mini Nasdaq-100 futures (NQ), 2006–2025

Executive Summary

Volatility is clearly predictable; for returns and candle direction no useful improvement was demonstrated

Problem Markets are hard to forecast, but are all their characteristics equally hard? For the next NQ futures candle we ask what machine learning can predict: the return, the direction or the realized volatility.

Volatility

PREDICTABLE

+40%

less error than "next = current" (DNN 128-64-32, 5 min): best model of the study

Also beats HAR-OLS (+6.8%); rise or fall right 72% (majority 51%); all 5 models beat persistence

Direction

MARGINAL

+0.69 pp

accuracy over always-the-majority-class: 50.59% vs 49.90% (DNN 256-128-64-32, 5 min)

About 7 extra right calls per 1,000 candles; seen in 2021–2023, gone in 2024–2025

Return

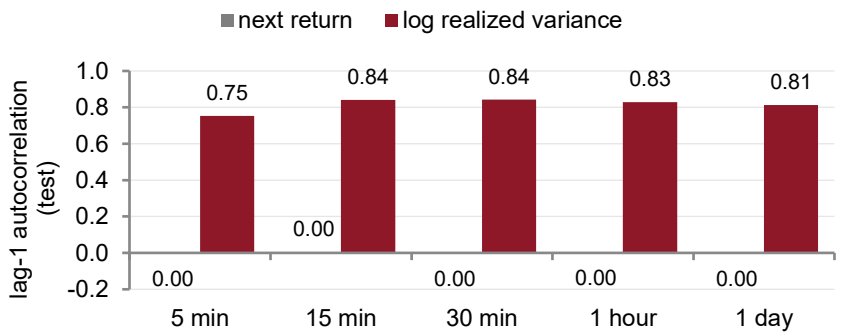
NOT DEMONSTRATED

R^2 −0.04%

vs simply forecasting zero, 95% interval [−0.10%, +0.02%] (DNN 256-128-64-32, 5 min)

Forecasts move by less than ±0.5 bp while a typical 5 min return is about 7.7 bp: the signal is lost in the noise

Why: volatility clusters, returns do not



Method

NQ futures 2006–2025, 5 candle lengths (5 min–1 day); 15 models, from benchmarks to deep networks and Mistral's Timestral. Train 2006–17, select 2017–21, test 2021–25.

Innovation

One pipeline, one data set and one test period for three different targets, so "what is predictable" is answered with the same models on every task and timeframe.

Usefulness

Volatility forecasts feed risk limits, position sizing and volatility-sensitive decisions. Direction and return forecasts at these horizons are too weak to act on.

Conclusion

Realized volatility is the most predictable characteristic; for returns and direction no useful improvement was demonstrated with the models, data and horizons tried.

Exploratory: the test period (2021-07 → 2025-05; daily to 2025-01) was examined in earlier project phases and the volatility selection score was revised after a first look at test results. Intervals: 95% bootstrap in 20-session blocks. pp = percentage points, bp = 0.01%.

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Data & Methodology

One instrument, five candle lengths, three targets, one chronological split

NASDAQ historical data
NQ 1-min, 2006–2025

→ **Feature engineering**
lags, momentum, volatility, time of day

→ **Models**
benchmarks · ML · DNN · Timestral

→ **Forecasts**
for candle t+1

→ **Out-of-sample evaluation**
test from 2021-07 (exploratory)

Three targets, all for the next candle (t+1), decided at the close of t

Target	Definition	Evaluated with
1 Next return	r = log(C(t+1)/C(t)), regression	MAE, RMSE, out-of-sample R^2 vs zero forecast
2 Next-candle direction	UP if C(t+1) > O(t+1); doji kept, counts as a miss	accuracy, balanced accuracy, F1 vs majority class
3 Future realized volatility	RV = Σ (1-min log returns)² inside candle t+1	log-RV error (R^2) vs persistence; RV UP if RV(t+1) > RV(t)

- **Instrument:** E-mini Nasdaq-100 futures (NQ, CME), back-adjusted continuous series. No QQQ, MNQ, cash index or VIX.
- **Split:** train 2006-01-10 → 2017-08-24, validation → 2021-07-14, test 2021-07-15 → 2025-05-30 (daily test ends 2025-01-30).
- **Leakage control:** features use candles ≤ t only; scaling fitted on training data; labels never cross a session or a split boundary.
- **Selection:** hyper-parameters and the model ranking use validation data only (slide 8); the volatility score was revised after a first look at test numbers, so not blind: results are exploratory.

Data profile per timeframe (test sample)

Candle	Rows: train · val · test	UP / DOWN / doji	Return s.d. (bp)	ACF return	ACF log RV
5 min	769k · 269k · 273k	50% / 48% / 1.8%	7.7	−0.005	**+0.75**
15 min	264k · 88k · 90k	51% / 48% / 1.0%	13.3	+0.003	**+0.84**
30 min	133k · 45k · 44k	51% / 48% / 0.8%	18.5	−0.002	**+0.84**
1 hour	65k · 22k · 22k	52% / 48% / 0.5%	26.2	−0.000	**+0.83**
1 day	1,991 · 822 · 642	54% / 46% / 0.0%	116.9	−0.001	**+0.81**

Reading: candle direction is close to a coin flip (UP 50%–54%); ACF (lag-1 autocorrelation; intraday pairs within a session) of returns is about zero (−0.005 to +0.003), while log realized variance is strongly persistent (0.75 to 0.84). Features (23): past returns, momentum, rolling volatility, range, body, wicks, RSI, moving-average gaps, relative volume, time of day, past log RV (1, 5, 22 candles, same slot on 5 prior sessions). Timestral sees the last 512 closes.

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All Models Tested

15 models on 5 timeframes: 150 model–timeframe–task configurations, all run on the same data

Family	Model	Method	Tasks
Baselines	**Zero return / training mean**	naive forecast	R
	Majority class	always the most frequent training class	D
	Persistence	RV(t+1) = RV(t)	V
	Rolling mean (22 candles)	average past variance	V
Statistical	**HAR-OLS**	linear regression on past log RV (1, 5, 22 candles, same time slot)	V
	Ridge regression	L2-penalised linear, α chosen on validation	R · V
	Logistic regression	L2-penalised, C chosen on validation	D
Machine learning	**Random forest**	200 trees, depth 8	R · D · V
	XGBoost	gradient boosting, depth 4, early stopping	R · D · V
Deep learning	**DNN 64-32-16 (original)**	feed-forward, ReLU, AdamW, early stopping	R · D · V
	DNN 64-32		R · D · V
	DNN 128-64-32		R · D · V
	DNN 256-128-64-32		R · D · V
Mistral AI	**Timestral-5M, frozen weights**	zero-shot quantiles of the next close from 512 past closes; median → return; direction = median vs last close (target: next candle's body)	R · D
	Timestral-5M + supervised calibration	q10–q90 spread → log RV via a linear map fitted on validation targets (on validation: fitted on earlier sessions only)	V

Notes

- **R** return · **D** direction · **V** volatility
- **Mistral's role:** Timestral is a time-series foundation model (local weights, no API, no training here). It is not a language model and it does not read news. Scored on 20,000 random validation and 20,000 test origins per timeframe (all daily ones).
- **Timestral outcome:** frozen weights beat no benchmark on any timeframe (return, direction); the calibrated volatility variant beat persistence only at 5 min. Timestral is the fifth model compared on slides 5–7; per-timeframe numbers in the report tables.
- **Considered, not run:** LSTM (the feature networks reach their early-stopping optimum within a few epochs). Timestral 28M / 123M / 209M: weights available locally, not tested in this study.
- **Not used:** Mistral chat / LLM API — no such integration exists in the repository.

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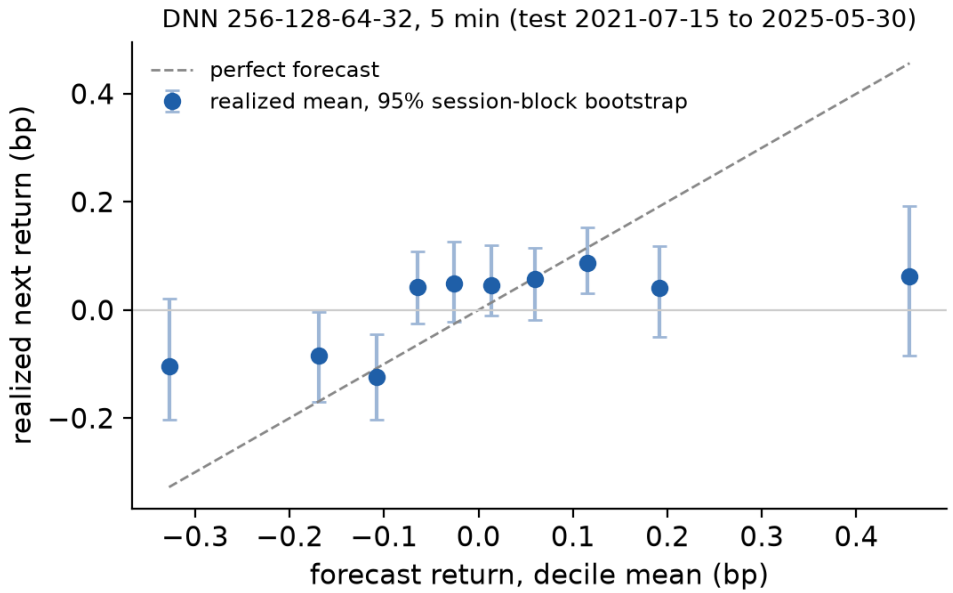
Price / Return Forecasting

Five models on 5 min candles: none improves on the zero-return forecast

#	Model	Val. z	MAE bp	RMSE bp	R^2 vs zero	95% interval (%)	Sign right
1	**DNN 256-128-64-32**	3.1	4.49	7.71	**−0.04%**	[−0.10, +0.02]	51.3%
2	**XGBoost**	2.5	4.50	7.71	**+0.00%**	[−0.02, +0.02]	51.2%
3	**Ridge**	0.8	4.50	7.71	**−0.03%**	[−0.09, +0.03]	51.2%
4	**Random forest**	0.1	4.50	7.71	**−0.05%**	[−0.09, −0.02]	51.2%
5	**Timestral-5M (frozen weights) †**	-4.3	4.62	7.91	**−2.55%**	[−3.43, −1.62]	50.2%

Benchmark (zero-return forecast): RMSE 7.71 bp on all 5 min test candles. † Timestral scored on its fixed random sample of 20,000 test candles (benchmark on the same candles).

- **Result:** the best R^2 is +0.00%; every interval includes or lies below zero, so no model beats "the next return is zero". A positive R^2 would mean a smaller squared error than that forecast.
- **Predicted vs realized (right, #1):** realized return per forecast decile: the forecasts span less than ±0.5 bp while a typical candle moves about 7.7 bp.
- **Reading:** classical, tree, deep and foundation models all fail in the same way: no usable return signal with the models, data and horizons tried (consistent with, but not proof of, a random walk).



Rows ranked by validation evidence z (improvement over the naive benchmark ÷ its 20-session block-bootstrap standard error); all other columns are test results. R^2 interval: model and zero-forecast errors resampled together. Test 2021-07-15 → 2025-05-30 (exploratory).

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Next-Candle Direction Classification

Five models on 5 min candles: 4 of 5 beat the majority class, by under one percentage point

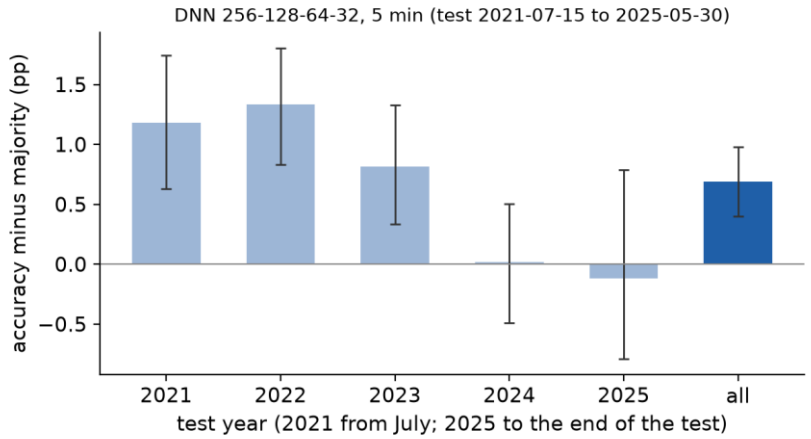
#	Model	Val. z	Accuracy	Balanced acc.	Majority	Gain	95% interval	Precision UP	Recall UP	F1 UP
1	**DNN 256-128-64-32**	11.9	**50.59%**	51.37%	49.90%	**+0.69 pp**	[+0.40, +0.98]	52.0%	60.4%	0.559
2	**Random forest**	11.0	**50.52%**	51.36%	49.90%	**+0.62 pp**	[+0.32, +0.91]	52.1%	56.3%	0.541
3	**Logistic**	11.0	**50.40%**	51.23%	49.90%	**+0.50 pp**	[+0.24, +0.77]	51.9%	56.7%	0.542
4	**XGBoost**	10.7	**50.47%**	51.34%	49.90%	**+0.58 pp**	[+0.27, +0.89]	52.1%	54.9%	0.535
5	**Timestral-5M (frozen weights) †**	-1.9	**49.25%**	50.10%	49.84%	**−0.59 pp**	[−1.65, +0.45]	50.8%	52.7%	0.517

† Timestral scored on its fixed random sample of 20,000 test candles (benchmark on the same candles). Accuracy counts dojis as misses.

#1 DNN 256-128-64-32: confusion matrix (test)

	Actual UP	Actual DOWN
Predicted UP	**82,434 (30.7%)**	76,180 (28.4%)
Predicted DOWN	53,956 (20.1%)	**55,837 (20.8%)**

- **Size:** +0.69 pp ≈ 7 more right calls per 1,000 candles. Tree, linear and deep models land within a few tenths of each other; Timestral does not beat the majority class.



Stability of #1 by test year: gain above zero in 2021, 2022, 2023; interval includes zero in 2024, 2025.

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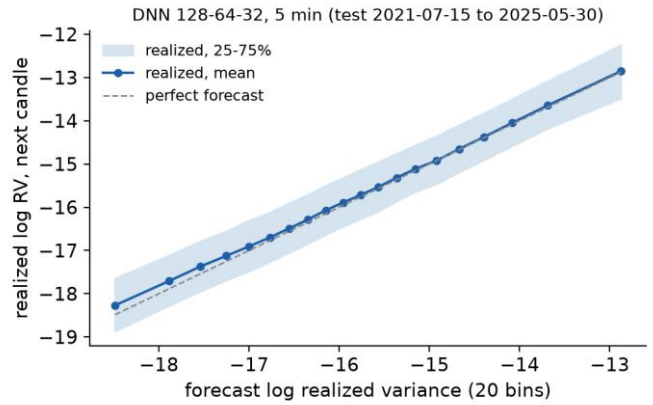
Volatility Forecasting

Five models on 5 min candles: every one beats persistence; the tree and deep models also beat HAR-OLS

#	Model	Val. z	RMSE log RV	R^2 vs persistence [95%]	R^2 vs HAR-OLS [95%]	RV up/down right	Balanced	RV dir. vs HAR-OLS [95%]
1	**DNN 128-64-32**	44.0	**0.901**	**+40.0% [+39.5, +40.4]**	**+6.8% [+6.2, +7.3]**	**72.2%**	72.0%	**+1.13 pp [+0.95, +1.31]**
2	**XGBoost**	42.4	**0.885**	**+42.0% [+41.5, +42.4]**	**+9.9% [+9.4, +10.3]**	**72.8%**	72.6%	**+1.71 pp [+1.52, +1.88]**
3	**Random forest**	27.4	**0.922**	**+37.1% [+36.5, +37.6]**	**+2.3% [+1.7, +2.8]**	**71.4%**	71.3%	**+0.37 pp [+0.25, +0.50]**
4	**Ridge**	23.1	**0.932**	**+35.7% [+34.9, +36.3]**	**+0.1% [−0.9, +0.9]**	**70.8%**	70.6%	**−0.27 pp [−0.51, −0.05]**
5	**Timestral-5M + calibration †**	6.4	**1.087**	**+12.2% [+9.7, +14.8]**	**−35.9% [−38.8, −33.2]**	**67.7%**	67.7%	**−3.36 pp [−4.02, −2.71]**

Benchmarks on all 5 min test candles: persistence (RV next = RV now) RMSE 1.163, majority 51.1% up/down; HAR-OLS RMSE 0.933, 71.0% up/down. † Timestral scored on its fixed random sample of 20,000 test candles (benchmark on the same candles).

- **Method:** realized variance (RV) of the next candle = sum of its squared 1-minute log returns; models forecast log RV. R^2 = share of the benchmark's squared log-RV error removed.
- **Reading:** persistence is naive; HAR-OLS already captures clustering and the time-of-day pattern, so the gain over HAR-OLS is what ML adds. On the test XGBoost (#2 on validation) is slightly ahead of #1 (+42.0% vs +40.0%).
- **Options (qualitative):** rising realized volatility favours long straddles / strangles, falling favours short ones; option prices embed IMPLIED volatility, so a correct realized-volatility forecast is not an options profit. Nothing on options was computed.



Rows ranked by validation evidence z (improvement over the naive benchmark ÷ its 20-session block-bootstrap standard error); all other columns are test results. Volatility score revised from QLIKE to log-RV error after a first look at test numbers: not blind.

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Which NASDAQ Market Characteristic Is Most Predictable?

Best model overall: DNN 128-64-32 forecasting 5 min realized volatility

How the ranking was made

1. Every model × timeframe is scored on VALIDATION data (2017–2021) by its improvement over the task's naive benchmark: zero return, majority class, persistence.
2. Evidence score z = that improvement ÷ its standard error (bootstrap in 20-session blocks). z is unit-free, so the three tasks can be compared.
3. Within each task the highest z wins (the five-model tables on slides 5–7). The best model overall is the task winner with the highest z.
4. Only then is it scored on the TEST block (2021–2025, exploratory). The test does not change the ranking.

Rank	Task	Task winner	Val. z	Test vs benchmark
1	**Volatility**	DNN 128-64-32, 5 min	**44.0**	+40.0% vs persistence [+39.5, +40.4]
2	**Direction**	DNN 256-128-64-32, 5 min	**11.9**	+0.69 pp vs majority [+0.40, +0.98]
3	**Return**	DNN 256-128-64-32, 5 min	**3.1**	R^2 −0.04% vs zero [−0.10, +0.02]

BEST MODEL OVERALL

DNN 128-64-32 · 5 min realized volatility

+40.0%

less squared log-RV error than persistence on the test, 95% interval [+39.5%, +40.4%]; validation z 44.0, 3.7× the direction winner

+6.8%
vs HAR-OLS [+6.2, +7.3]

72.2%
RV up/down right (majority 51.1%, HAR 71.0%)

0.901
log-RV RMSE (persistence 1.163, HAR 0.933)

Honest caveat: on the test XGBoost (#2 on validation) is slightly better (+42.0%); the ranking follows the validation rule fixed in advance, not the test.

Exploratory verdict: realized volatility is the most predictable characteristic; for returns no improvement was demonstrated, and the direction gain is small and fades after 2023. **Originality** three targets, five timeframes, classical to foundation models in one pipeline. **Usefulness** volatility forecasts for risk limits and sizing. **Feasibility** laptop CPU, free local models, reproducible (next slide).

Laboratory conditions: clean historical data, no commissions, spread, slippage or latency. In live markets these apply, so the values shown are not reproducible in practice and a small edge may disappear.

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Reproducibility

All tables, charts and slides are regenerated from the code in the repository; the licensed NQ data are not published

Code, results and this deck https://github.com/Paciooo/Project_ai_ML_bigdata · branch main

Step	What	Command (PowerShell, repository root)
1 Environment	Python 3.12, Node.js 18+	`pip install -r requirements.txt -r requirements-timestral.txt`
2 Data (not in Git)	NQ 1-minute bars 2006–2025 (TradeStation, licensed); Timestral-5M weights from the local Hugging Face cache	`$env:FUTURES_DATA_DIR = "C:\path\to\futures-data"`
3 Train and forecast	all models, five timeframes, CPU only	`python multitarget.py --tag final`
4 Tables and charts	results_*.csv, data_profile.csv, selection, charts	`python multitarget_report.py <run> reports\multitarget_final`
5 Independent check	recomputes every number from the saved forecasts	`python verify_multitarget.py <run> reports\multitarget_final presentation`
6 Tests and deck	unit tests; slides from deck_data.json	`python -m pytest -q · cd presentation; npm ci; node build_deck.js`

What has been checked

- verify_multitarget.py: every metric, the selection rule, the data profile and the deck data recomputed independently; 0 problems on the final run
- unit tests for targets, bootstrap, selection rule and the causal Timestral calibration
- run 20261112T050812Z_82953a_final

Limits

- **Laboratory conditions:** live trading adds commissions, spread, slippage and latency: these values are not reproducible in practice and a small edge may disappear
- the test period had been examined before: results are exploratory
- without the licensed NQ data the training cannot be re-run
- run made from a working copy; run.json stores source-file hashes

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